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CREDIT RISK PREDICTION AND DEFAULT ANALYTICS SYSTEM

Course: Data analytics

Submitted By:

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Abstract

Credit risk assessment is a crucial process in the banking and financial sector to determine the likelihood of a borrower defaulting on a loan. Accurate prediction of credit risk helps financial institutions minimize losses and make informed lending decisions. Traditional credit evaluation methods often rely on manual analysis and rule-based systems, which may not effectively handle large and complex datasets.

This project proposes a using **Machine learning techniques** to predict loan default probability. A **Logistic Regression model** was developed using **Python and the Scikit-Learn library** to analyze borrower attributes such as **income, age, loan amount, loan intent, and credit grade**. The model was trained on historical loan data and evaluated using performance metrics including **accuracy, confusion matrix, and ROC-AUC score**.

To provide meaningful business insights, an **interactive Power BI dashboard** was designed to visualize key credit risk indicators such as **total customers, default rate, loan distribution, and risk segmentation**. Based on the predicted probabilities, borrowers are categorized into **Low Risk, Medium Risk, and High Risk groups**, enabling financial institutions to identify high-risk applicants and improve credit decision-making.

The proposed system demonstrates the effectiveness of combining **machine learning and data visualization** to enhance credit risk management and support data-driven financial decision processes.

TABLE OF CONTENTS

CONTENT	PAGE.NO
1.Introduction • Project Background • Problem Statement • Objectives of the Study	**4-5**
2. Tools and Technologies • Python • Pandas and NumPy • Scikit-Learn • Matplotlib and Seaborn • MySQL • Power BI • Streamlit	**5**
3. System Architecture	**6**
4. Data Description • Dataset Overview • Features and Variables	**7**
5. Exploratory Data Analysis (EDA) • Loan Status Distribution • Loan Amount Distribution • Correlation Analysis • MySQL Data Analysis Summary	**8-11**
6. Feature Engineering and Data Preprocessing • Data Cleaning • Feature Creation • Encoding and Data Preparation	**11**
7. Methodology Diagram	**12**
8. Predictive Modeling • Linear Regression Model • Logistic Regression Model • Random Forest Model	**13**

9. Model Evaluation and AI Implementation • Accuracy Score • Confusion Matrix • ROC Curve • Feature Importance	**14-16**
10. Dashboard and Interactive Visualization • Power BI Dashboard • Streamlit Application	**16-17**
11. Business Insights and Findings	**18**
12. Conclusion	**18**
13. Future Scope	**18**
14. References	**19**

1. Introduction

Credit risk assessment is an important process in the banking and financial sector to evaluate whether a borrower can repay a loan. Credit risk refers to the possibility that a borrower may fail to meet their financial obligations, resulting in financial losses for lending institutions.

With the growth of digital banking, financial institutions handle large volumes of customer data. Traditional credit evaluation methods based on manual analysis are often slow and less efficient.

This project develops a **Credit Risk Prediction System using machine learning** to predict the probability of loan default. A **Power BI dashboard** is also created to visualize loan performance, default trends, and borrower risk categories, helping financial institutions make **data-driven lending decisions**.

❖ Project Background

Credit risk management is a crucial function in the banking and financial sector. Financial institutions must carefully evaluate borrowers before approving loans to reduce the risk of financial losses.

Traditional credit risk assessment methods rely on **credit scores, manual verification, and rule-based systems**, which may not effectively analyze large and complex datasets.

With the advancement of **machine learning and data analytics**, predictive models can analyze historical loan data to estimate the probability of loan default. This project applies machine learning techniques to develop a system that **identifies high-risk borrowers and provides valuable insights for better lending decisions**.

❖ Problem Statement

Financial institutions face significant challenges in identifying high-risk borrowers before approving loans. Incorrect lending decisions can result in loan defaults, financial losses, and increased operational risks.

The primary problem addressed in this project is:

How can machine learning techniques be used to accurately predict loan defaults and help financial institutions reduce credit risk?

To solve this problem, this project develops a predictive model that analyzes borrower data and classifies customers based on their risk level.

❖ Objectives of the Study

The main objectives of this project are:

1. To analyze borrower data and identify key factors influencing loan defaults.
2. To build a machine learning model that predicts loan default probability.
3. To categorize borrowers into risk groups such as low risk, medium risk, and high risk.
4. To evaluate model performance using accuracy, ROC-AUC score, and confusion matrix.
5. To develop an interactive Power BI dashboard for visualizing credit risk insights.

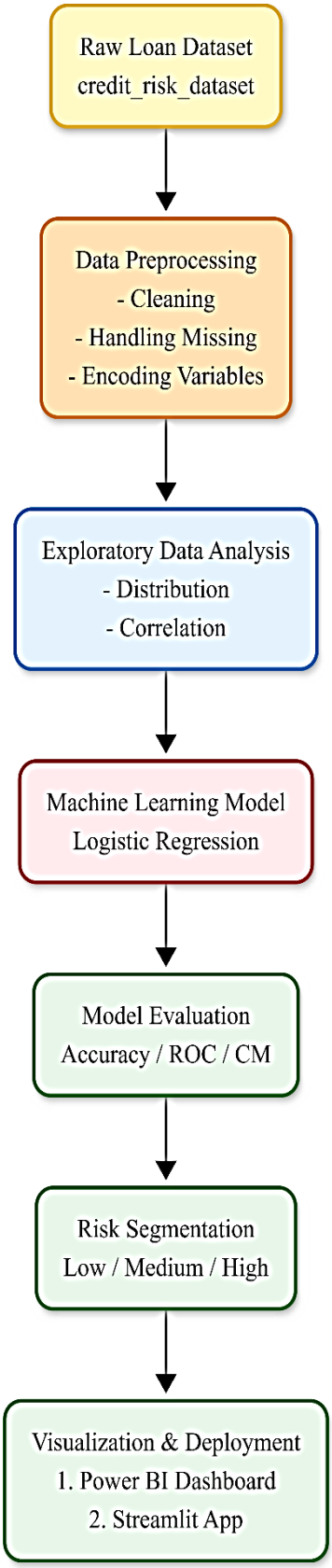
2. Tools and Technologies

This project uses various tools and technologies for **data analysis, model development, and visualization**.

- **Python**
  Python is the primary programming language used for data analysis and machine learning model development.
- **Pandas**
  Pandas is used for **data manipulation, cleaning, and preprocessing**.
- **NumPy**
  NumPy provides support for **numerical computations and array operations**.
- **Scikit-Learn**
  Scikit-Learn is used to implement **machine learning algorithms such as Logistic Regression** and perform model evaluation.
- **Matplotlib and Seaborn**

  These libraries are used for **data visualization and exploratory data analysis (EDA)**.
- **MySQL**
  MySQL is used for **storing and managing the dataset in a structured database format** and for performing data queries.
- **PowerBI**
  Power BI is used to create **interactive dashboards and visualizations** for credit risk insights and loan analysis.
- **Streamlit**
  Streamlit is used to build a **web application that allows users to predict loan default probability interactively**.

3. System Architecture



4. Data Description

❖ Dataset Overview

The dataset used in this project contains historical loan records of borrowers. Each record represents a customer along with their financial attributes and loan details.

The dataset includes features such as:

- Age of the borrower
- Annual income
- Loan amount
- Loan intent
- Home ownership status
- Loan grade
- Default status

The **target variable** in the dataset is **loan_status**, which indicates whether a borrower defaulted on the loan.

❖ Features and Variables

Some of the important variables in the dataset include:

person_age
Age of the borrower.

person_income
Annual income of the borrower.

person_home_ownership
Indicates whether the borrower owns, rents, or mortgages their home.

loan_amnt
Amount of loan taken by the borrower.

loan_intent
Purpose of the loan.

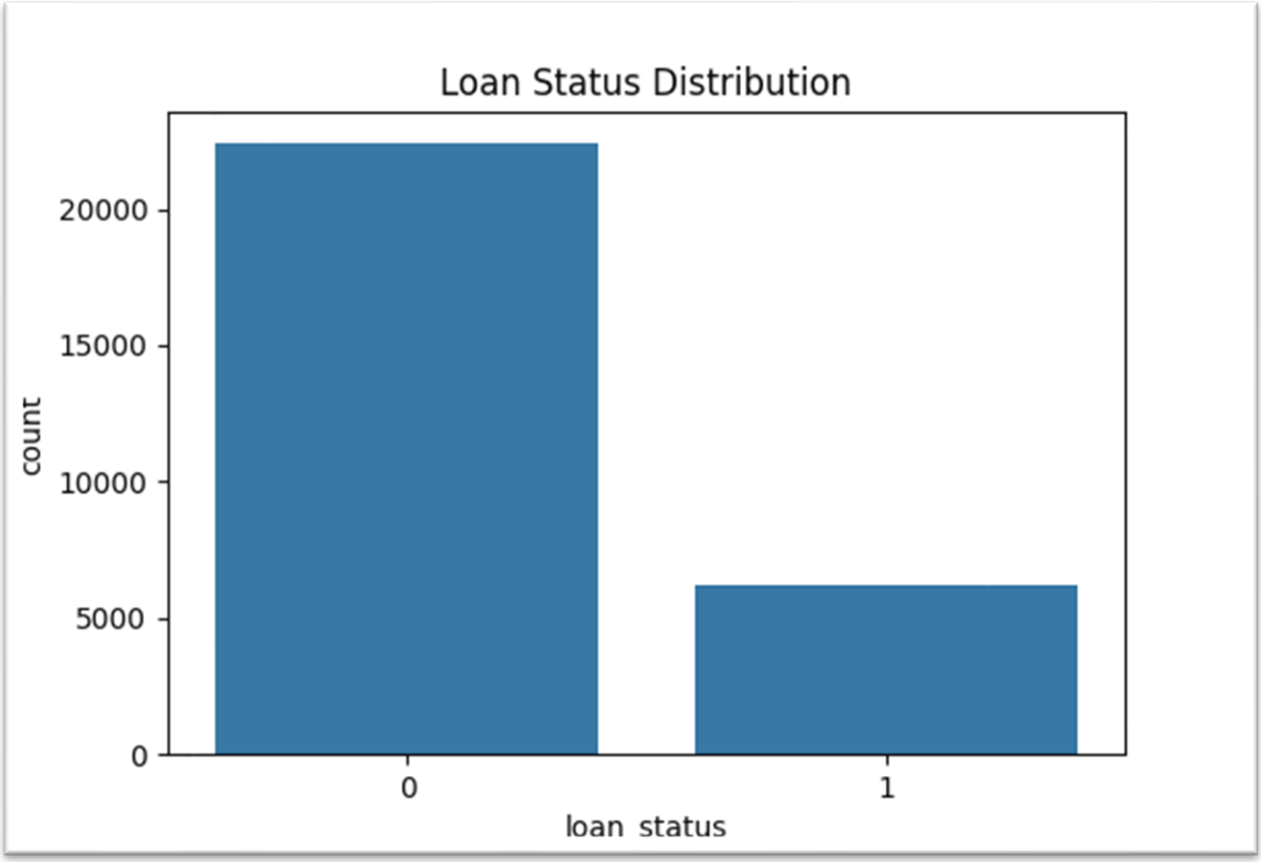
loan_grade
Credit grade assigned to the loan.

loan_status
Target variable indicating loan default.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis helps understand the dataset and identify patterns.

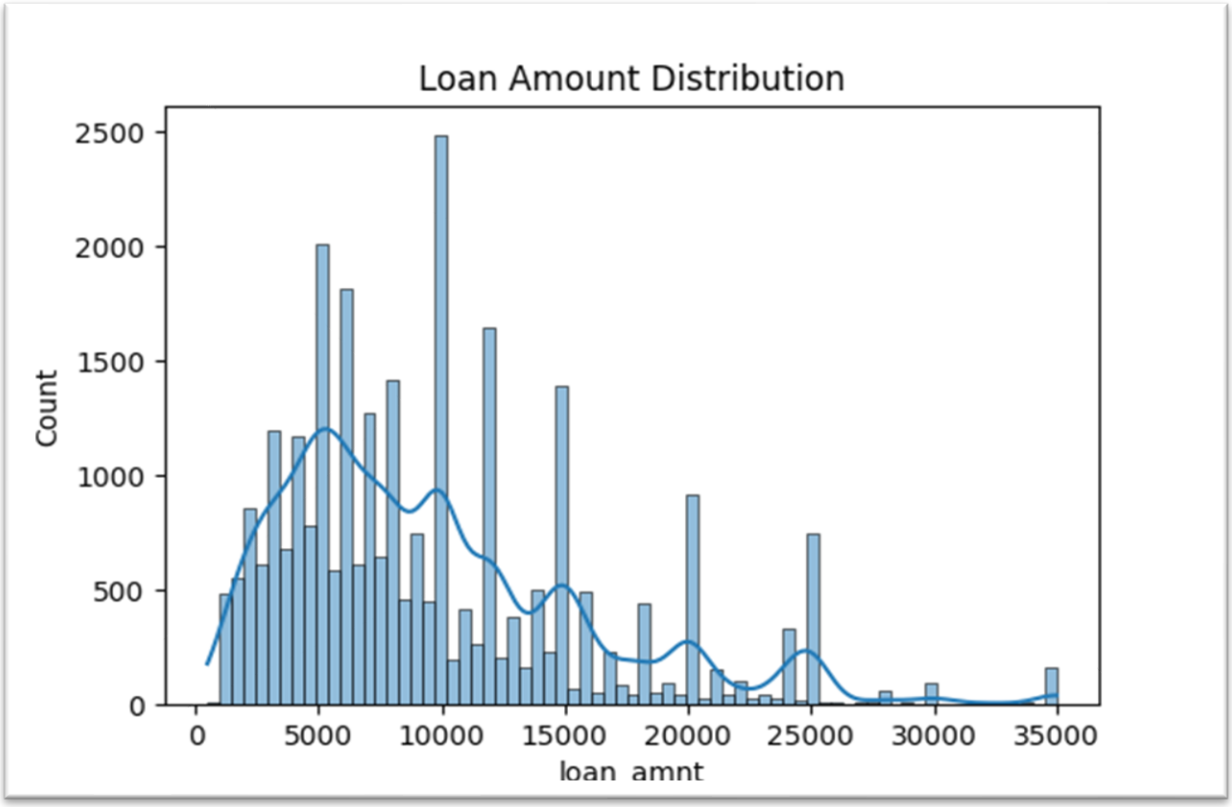
❖ Loan Status Distribution



This chart shows the distribution of borrowers based on **loan repayment status**.
Value **0 represents non-default borrowers**, while **1 represents default borrowers**.

The visualization indicates that **most borrowers successfully repay their loans**, while a smaller portion defaults. This imbalance in the dataset highlights the importance of using machine learning models that can accurately detect **high-risk borrowers**.

❖ Loan Amount Distribution



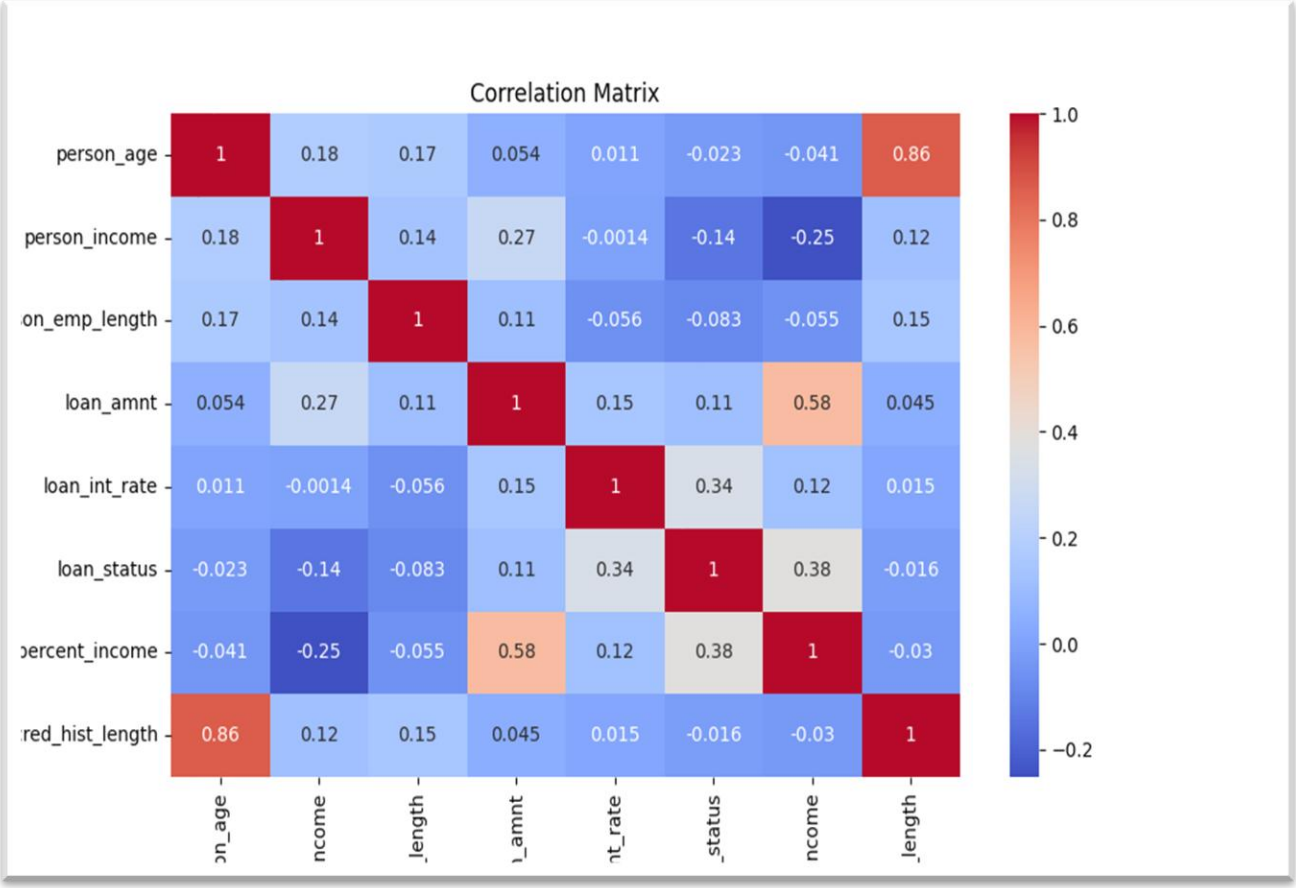
The histogram visualizes the **distribution of loan amounts issued to borrowers**.

The analysis shows that:

- Most borrowers take **small to medium-sized loans**.
- Only a small number of borrowers receive **very large loans**.

Understanding loan amount distribution helps financial institutions analyze **lending patterns and potential financial exposure**.

❖ Correlation Heatmap



The correlation matrix displays the **relationship between numerical variables** in the dataset. It helps identify which features are positively or negatively related to each other and to the **loan default status**.

From the analysis:

- **Loan interest rate** and **loan percent income** show a moderate relationship with loan default.
- **Loan amount** is positively correlated with **loan percent income**, indicating that larger loans often represent a higher portion of borrower income.
- **Person age and credit history length** show a strong positive correlation, meaning older individuals generally have longer credit histories.

This analysis helps in **feature selection for building effective machine learning models**.

❖ MySQL Data Analysis Summary

- MySQL was used to **store and manage the credit risk dataset** in a structured database.
- SQL queries such as **SELECT, COUNT, and DESCRIBE** were used to explore the dataset and understand its structure.
- Queries were performed to analyze **loan default distribution and borrower characteristics**.
- Default rates were calculated based on **loan grade, home ownership, and loan intent**.
- SQL analysis helped identify **high-risk patterns related to income, loan amount, and interest rate**.
- Credit history analysis was conducted to evaluate **default risk across different credit history lengths**.
- A **SQL view (default_summary)** was created to summarize default rates by loan grade for **Power BI dashboard integration**.

6. Feature Engineering and Data Preprocessing

Feature engineering and data preprocessing are important steps used to prepare the dataset for machine learning analysis. These steps help improve **data quality, model performance, and prediction accuracy**.

❖ Data Cleaning

Data cleaning was performed to ensure the dataset was **accurate and consistent**. Missing values were identified and removed, and data types were checked to ensure the dataset was suitable for analysis.

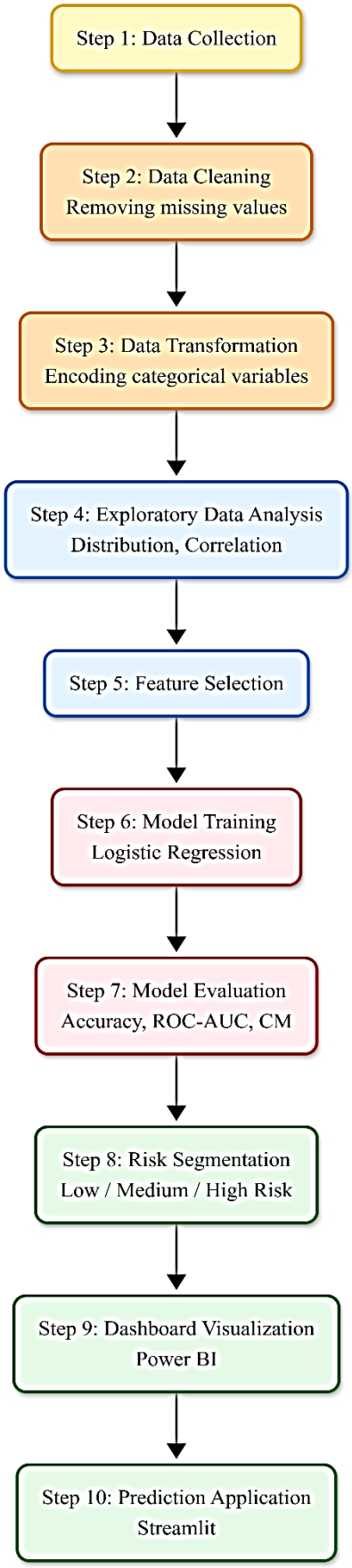
❖ Feature Creation

New features were created to improve the analysis. A **default probability score** was generated using the machine learning model, and borrowers were categorized into **Low Risk, Medium Risk, and High Risk groups** based on their predicted default probability.

❖ Encoding and Data Preparation

Categorical variables such as **loan intent, loan grade, and home ownership** were converted into numerical format using **one-hot encoding**. The dataset was then divided into **feature variables (X)** and the **target variable (loan_status)** for machine learning model training.

7. Methodology Diagram



8. Predictive Modeling

Predictive modeling was performed to develop machine learning models capable of **predicting the probability of loan default**. Different algorithms were explored to analyze borrower data and classify customers based on their risk level.

❖ Linear Regression Model

Linear Regression was initially considered to analyze the relationship between variables. However, since loan default prediction is a **binary classification problem**, regression models are not the most suitable for final prediction.

❖ Logistic Regression Model

Logistic Regression was used as the **primary model for predicting loan default probability**. It is widely used for binary classification problems where the output has two possible outcomes, such as **default or non-default**.

The dataset was divided into:

- **80% Training Data** for training the model
- **20% Testing Data** for evaluating model performance

The model was trained on the training dataset and then tested on unseen data to measure its prediction accuracy.

❖ Random Forest Model

Random Forest is an ensemble machine learning algorithm that uses multiple decision trees to improve prediction accuracy. This model helps capture **complex patterns and relationships within the data**, making it effective for credit risk prediction.

Using multiple models helps compare performance and select the most suitable approach for predicting **borrower default risk**.

9. Model Evaluation and AI Implementation

The trained machine learning model was evaluated using several performance metrics to measure its ability to correctly predict loan default risk.

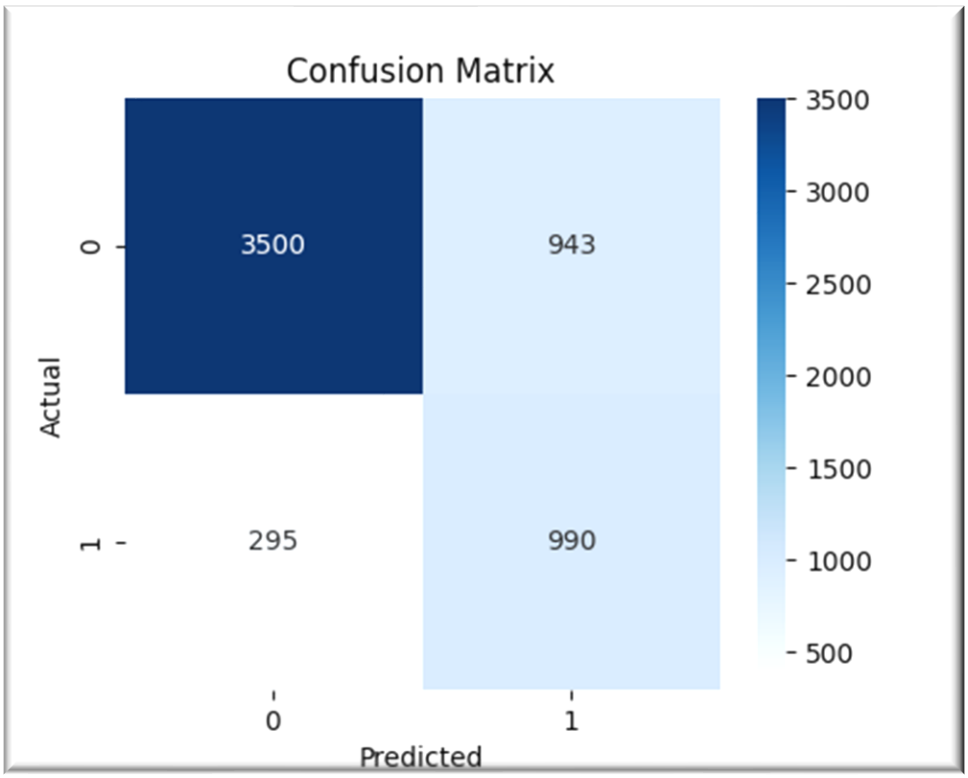
❖ Accuracy Score

Accuracy measures the proportion of correct predictions made by the model. The Logistic Regression model achieved an **accuracy of 78.38%**, indicating that the model correctly predicted most loan outcomes in the testing dataset.

❖ ROC–AUC Score

The **ROC–AUC score** evaluates the model's ability to distinguish between default and non-default borrowers.
The model achieved a **ROC–AUC score of 0.85**, which indicates good classification performance.

❖ Confusion Matrix



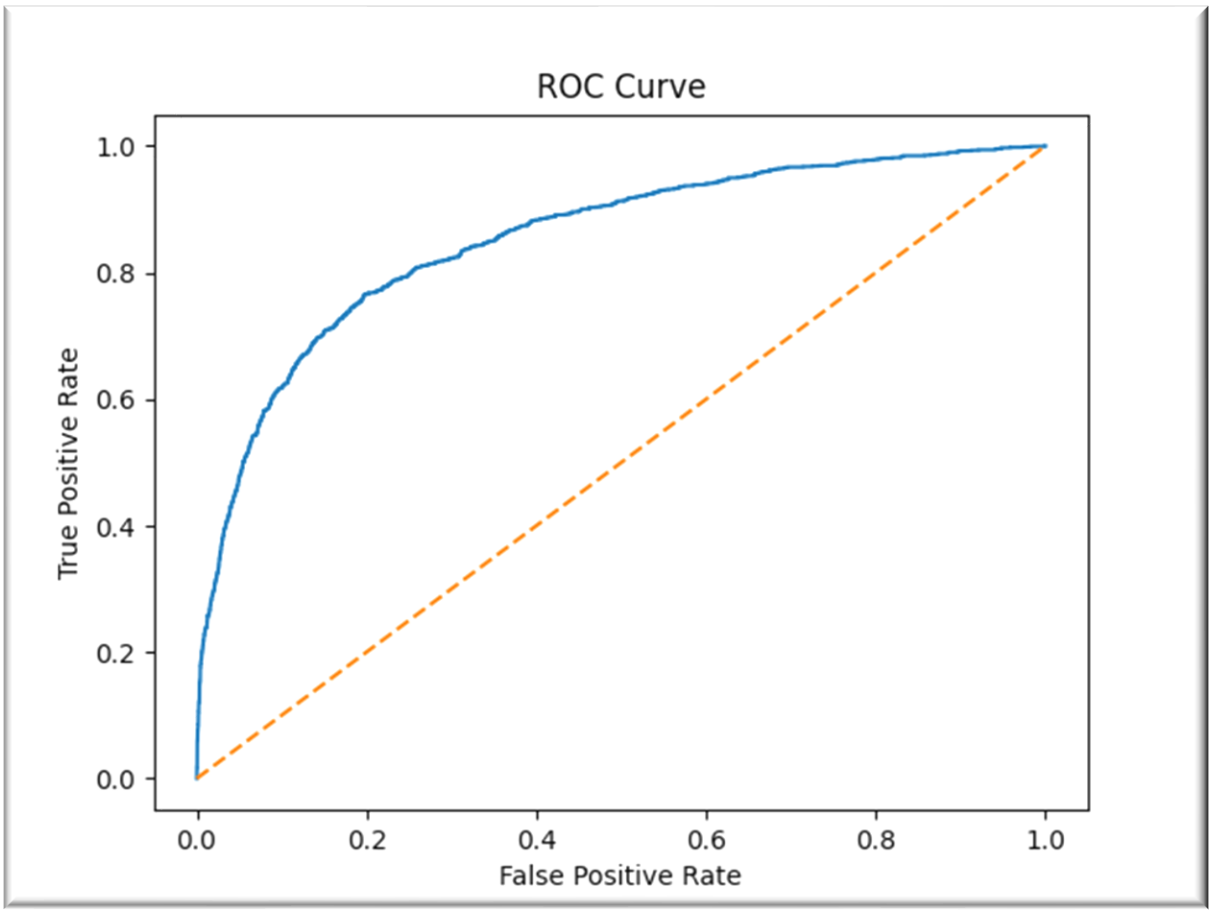
The confusion matrix shows the performance of the classification model by comparing **actual values with predicted values**.

- **3500 True Negatives (TN):** Borrowers who did not default and were correctly predicted as non-default.

- **990 True Positives (TP):** Borrowers who defaulted and were correctly identified by the model.
- **943 False Positives (FP):** Borrowers predicted as default but actually repaid their loans.
- **295 False Negatives (FN):** Borrowers predicted as non-default but actually defaulted.

This analysis helps evaluate the **model's ability to correctly identify risky borrowers and reduce lending risk**.

❖ ROC Curve



The ROC (Receiver Operating Characteristic) curve illustrates the model's ability to **distinguish between default and non-default borrowers**.

The curve plots the **True Positive Rate (TPR)** against the **False Positive Rate (FPR)** at different classification thresholds. The model achieved a **ROC-AUC score of approximately 0.85**, indicating strong classification performance.

Since the ROC curve is **closer to the top-left corner**, it demonstrates that the model effectively separates **high-risk borrowers from low-risk borrowers**, making it suitable for credit risk prediction.

❖ Feature Importance

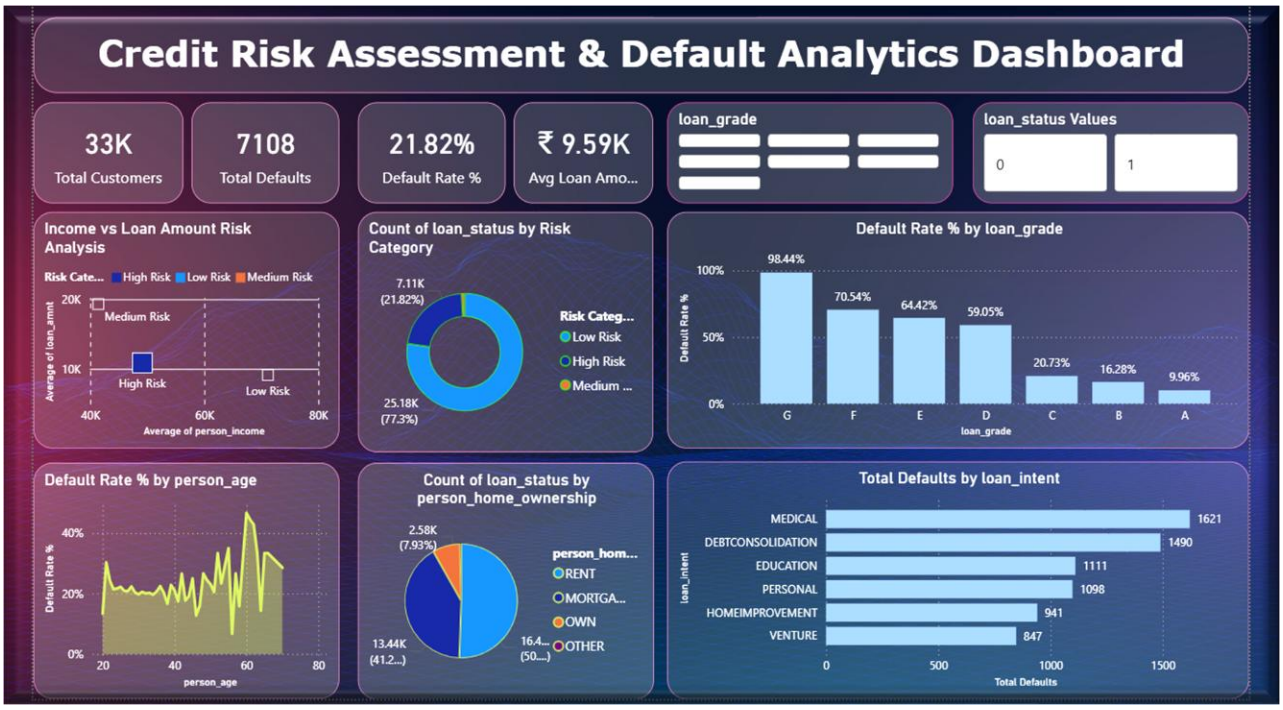
Feature importance analysis was performed to understand which variables had the greatest influence on loan default prediction. Key contributing factors include:

- **Loan percent income**
- **Loan grade**
- **Home ownership status**
- **Interest rate**
- **Previous credit default history**

These features play a significant role in determining the borrower's credit risk level.

10. Dashboard and Interactive Visualization

❖ Power BI Dashboard



This dashboard analyzes loan data to understand **borrower risk and loan default patterns**. The dataset contains **33K customers**, with about **21.8% default rate**.

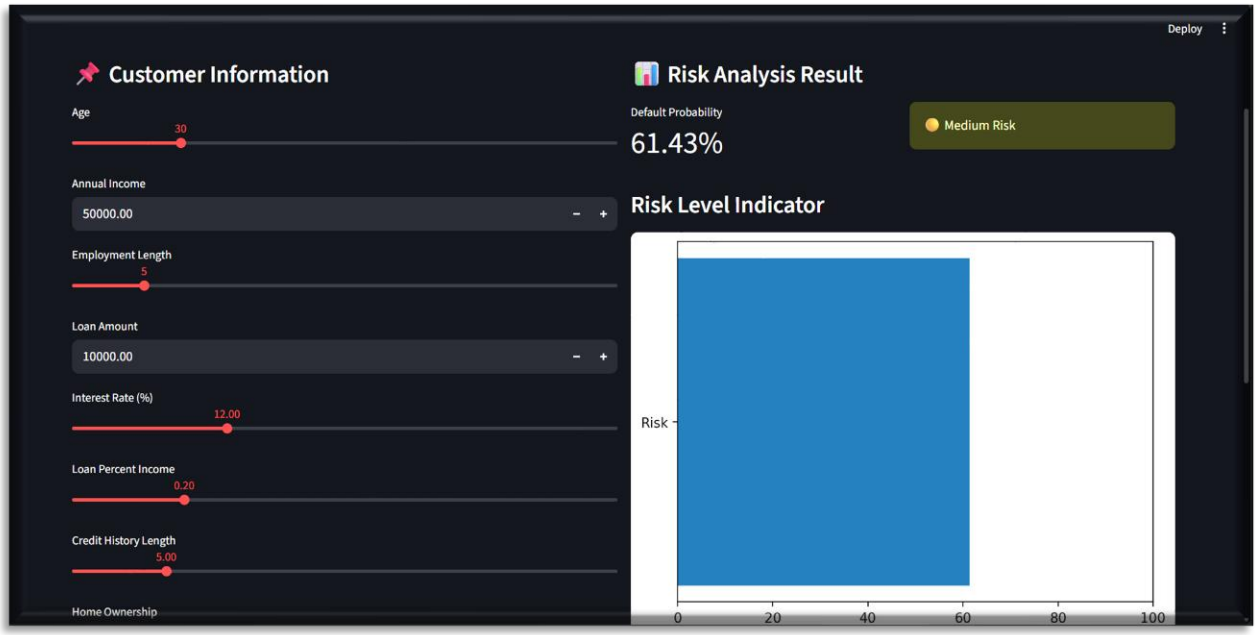
Key visuals show:

- **Income vs Loan Amount** to identify **low, medium, and high-risk borrowers**.
- **Default Rate by Loan Grade**, where lower grades (F, G) have higher defaults.
- **Default by Age and Loan Purpose** to understand which groups default more.

Overall, the dashboard helps lenders **identify risky borrowers and make better loan approval decisions.**

❖ Streamlit Application

- **Credit Risk Prediction Interface**



This interactive **Streamlit application** predicts the **loan default risk of a customer** based on inputs like age, income, employment length, loan amount, interest rate, and credit history.

The system uses a **machine learning model** to calculate the **default probability** and classify the borrower into **Low, Medium, or High Risk**.

It helps financial institutions **quickly assess customer creditworthiness and make better loan approval decisions.**

11. Business Insights and Findings

- **Loan grade is a major factor in default risk** — borrowers with lower grades (F and G) show significantly higher default rates.
- Borrowers with a **high loan-to-income ratio** are more likely to face repayment difficulties and default on loans.
- **Higher interest rates** are often associated with higher-risk borrowers and increased default probability.
- Customers with **longer credit history and stable home ownership** generally show better repayment behavior.
- Certain loan purposes, such as **medical expenses and debt consolidation**, have relatively higher default trends.

12. Conclusion

- This project developed a Credit Risk Prediction System using machine learning techniques to identify borrowers who are likely to default on loans. Using Logistic Regression, the model achieved an accuracy of 78.38% and a ROC-AUC score of 0.85, indicating good prediction capability.

- The integration of Python for model development, MySQL for data storage, Power BI for visualization, and Streamlit for an interactive prediction interface provides a complete system for analyzing credit risk. This system helps financial institutions identify high-risk borrowers, reduce financial losses, and make more informed lending decisions.

13. Future Scope

- Implement **advanced machine learning models** such as Random Forest, XGBoost, or Gradient Boosting to improve prediction accuracy.
- Integrate **real-time banking or financial data** to allow instant credit risk evaluation during loan applications.
- Use **Explainable AI techniques (SHAP, LIME)** to make model predictions more transparent and understandable.
- Deploy the system on **cloud platforms** to make the application scalable and accessible for real-world financial institutions.

14. References

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6. Kaggle Dataset Repository – Credit Risk and Loan Default datasets used for machine learning practice.